Uncertainty Quantification and Sensitivity Analysis in a Nonlinear Finite-Element Model of a Permanent Magnet Synchronous Machine

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***Abstract*—The objective of this article is to set forth a computationally efficient methodology to quantify the effects and assess the relevance of geometric and material uncertainty on a permanent magnet synchronous machine (PMSM), based on a nonlinear finite-element model (FEM). Our methodology follows the theory of Gaussian process regression and principal component analysis to build a computationally inexpensive surrogate model that replaces the FEM. Uncertainty quantification and sensitivity analysis studies focus on the electromagnetic torque, flux linkage, and core loss of the PMSM.**

***Index Terms*—Finite-element analysis, Gaussian processes, permanent magnet machines, principal component analysis, sensitivity analysis, uncertain systems.**

I. Introduction

THE manufacturing process of an electric machine may cause a noticeable deviation from its nominal output. This is due to the lamination punching process, the raw materials used, and various other factors that cannot be tightly controlled in a mass manufacturing environment [1]–[4]. Having a good understanding of the impact of these uncertainties is a prerequisite for robust design of high-performance machines, such as the ones used in electric transportation applications. In this work, we focus on a permanent magnet synchronous machine (PMSM). The PMSM is analyzed using a two-dimensional finite-element model (FEM), because it is ideal for modeling minute geometric and material variability around nominal design specifications.

The main contribution of this paper is a computationally efficient uncertainty quantification (UQ) methodology to characterize the effects on PMSM quantities of interest (QoIs). This is conducted in conjunction with a sensitivity analysis (SA), aiming to identify the most relevant uncertain parameters. UQ [5] and SA [6] typically depend on Monte Carlo sampling [7], which requires a large number of model evaluations to achieve convergence of the statistics. Therefore, directly employing computationally expensive, but high-fidelity, nonlinear PMSM FEMs is intractable.

The first ingredient of the proposed UQ/SA method is an appropriate stochastic input model. We consider three types of uncertainties. The first type is related to geometric features of the device. Deviations from nominal dimensions are caused by manufacturing tolerances and by the lamination punching process, due to the variable condition of the punching die [2]. The second type is related to the magnetic properties of the materials, namely, those of steel and the permanent magnets. This includes the effects of punching, which degrades the material close to the cut edge [8], [9]. The third type is related to the core loss. The underlying input space, considering uncertain and non-uncertain parameters, could have tens of independent parameters. Key aspects of this model are presented in Sections II-A and II-B. In particular, Section II-A presents important details of the PMSM under consideration, and Section II-C discusses the FEM output.

The second ingredient of the proposed UQ/SA method is a surrogate PMSM model. We construct an inexpensive analytical approximation of the FEM response surface using a relatively low number of input-output pairs provided by the high-fidelity FEM. To this end, we employ Gaussian process (GP) regression models, which have shown outstanding performance as emulators of physical systems, because they can represent a wide range of functions, while their Bayesian nature naturally yields the predictive uncertainty [10].

We construct the FEM surrogate with the following training process, which is detailed in Sections II-D and II-E. First, we generate a random set of FEM input-output observations. Second, we divide this dataset into training and validation counterparts. Third, for FEM outputs that are waveforms (i.e., torque or flux linkage as a function of rotor position), we use principal component analysis (PCA) [11] to reduce the dimensionality of the output space further (here, from hundreds of waveform samples to a few PCA coefficients). Fourth, we use GP regression (GPR) to learn the map between the FEM inputs and the reduced output (or the output itself in the case of scalar quantities, i.e., eddy current, excess and hysteresis losses). Fifth, we apply the surrogate model on the validation inputs. We ensure that the validation error is below a desirable threshold by increasing

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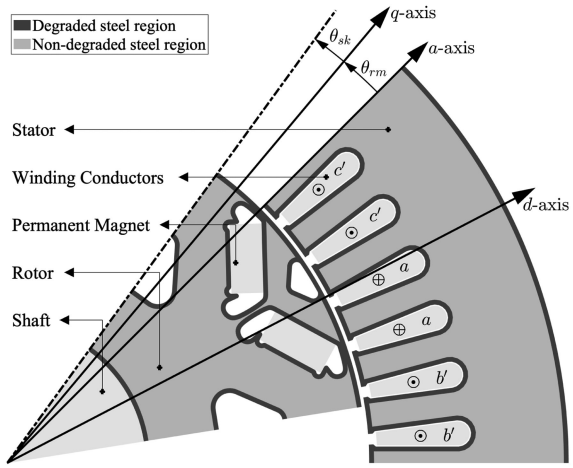


Fig. 1. Cross-section of one pole of the PMSM under study.

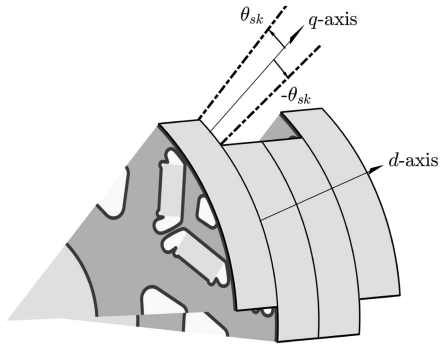


Fig. 2. Rotor assembly of the PMSM under study.

the size of the training dataset iteratively. The resulting surrogate functions as follows: it passes the inputs through the GPR to obtain PCA coefficients (or the scalar output), which are then passed through the inverse PCA transform to reconstruct the output waveform. We then use the PMSM surrogate to perform UQ and SA. Section III presents a case study. Specifically, we consider the effects of uncertainty on average torque, sixth torque harmonic magnitude, peak line-to-line voltage, and total core loss.

II. Proposed Methodology

A. PMSM Under Study

The device under study is an eight-pole PMSM, rated for 235 Nm and 96 kW at 3900 rpm. Its cross-section is depicted in Fig. 1, where θ_{rm} is the mechanical rotor angle.

The rotor lamination stack is divided into four identical sub-stacks, such that the rotor stack length equals the stator stack length, $\ell_m = 90$ mm. During assembly, two of the four sub-stacks are rotated clockwise, whereas the other two are rotated counterclockwise, as shown in Fig. 2. A skew angle, θ_{sk}, is defined to parameterize this rotation. Note that Fig. 1 shows the cross-section of a sub-stack with a positive skew angle. It should be noted that the *qd*-axes are representative of an average magnetization direction from the combined effect of all four sub-stacks. To analyze sub-stacks, first we develop a parametric two-dimensional FEM. This high-fidelity FEM provides input-output observations that are used to construct a computationally inexpensive surrogate model that eventually replaces the FEM. To account for the skewing of the rotor, we combine two surrogate model results, obtained for positive and negative skew angles, respectively.

We use *ANSYS Electronics Desktop/Maxwell* as the central simulation engine [12]. A custom Python-based interface was developed to enable the automatic generation of PMSM models for any given parameter vector. The software is deployed on a computational cluster, so that simulations can be parallelized.

B. Input Space Variability

There are scores of uncertain PMSM parameters the effects of which one could potentially study; however, the number that can be realistically incorporated in a single UQ study should be limited. The process we followed to narrow down the list of parameters was as follows. First, we compiled a list of all geometric, material, and operating parameters that either we believed would be important a priori or the effect of which was unknown. This list contained 27 parameters in total. Then, we eliminated all parameters the values of which depended on other parameters, e.g., the air gap, and all parameters the effect of which could be studied a posteriori, e.g., the rotor stack length. Finally, we eliminated all parameters that would break the symmetry of the PMSM, e.g., the rotor lamination centering tolerance. Even though our methodology could have readily handled these symmetry breaking parameters, the computational cost of generating the simulations would have increased from approximately one week to several months. After this elimination process, the number of remaining parameters was 18. The variability of each input parameter is representative of the expected range in a mass production line, as determined by the motor manufacturer. We classify the input parameters into four categories: 1) geometric, 2) material magnetic properties, 3) material loss properties, and 4) operating point.

1) Geometric Parameters: Geometric variability is caused by manufacturing tolerances, the lamination punching process due to the variable condition of the punching die, and the assembly process [2], [13], [14]. The subset of geometric parameters that we treat as uncertain contains the ones that we consider as most relevant to the PMSM performance.

Two key uncertain parameters are the inner stator diameter d_{is} and the outer rotor diameter d_{or}. Furthermore, the shape of the permanent-magnet (PM) pocket and the dimensions of the PMs are varied, identically for each pole, as shown in Fig. 3. We vary the pocket by moving its edge in a normal direction from the nominal shape. We use the signed distance γ as a parameter, representing the distance from the nominal edge (obtained for $\gamma = 0$), such that when the edge is displaced inwards or outwards, the signed distance is negative or positive, respectively. The width and thickness of a PM are denoted by w_{pm} and h_{pm}, respectively. We also consider the uncertainty about the skew angle θ_{sk}, which is caused by the assembly process of the sub-stacks on the shaft.

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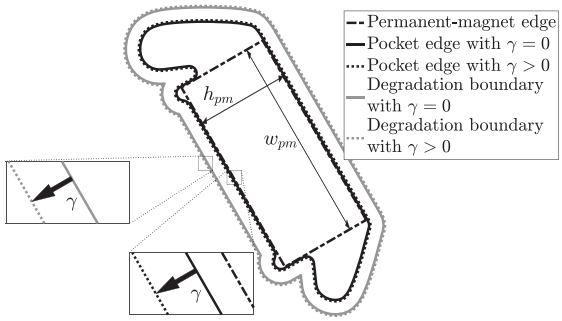


Fig. 3. Permanent-magnet and pocket geometry parameterization.

2) Parameterization of Material Magnetic Properties: The effects in PMSM performance due to the variability in the material properties can be significant. The main factor for the PM material is the uncertainty about the residual flux density, B_r, which can be traced to the PM manufacturing process and the quality of the magnet material, among other factors [13], [14]. On the other hand, the representation of the uncertain magnetic properties of steel requires a more sophisticated model of the *B-H* characteristic and its spacial variability within the machine.

The stator and rotor regions are divided into two sub-regions, as depicted in Fig. 1: (i) the degraded steel regions (dark shaded) close to the edges of the lamination, and (ii) the non-degraded steel regions (lightly shaded). The non-degraded steel properties may vary due to the grain structure uncertainty of the material, caused by the manufacturing process of non-oriented electrical steel sheets [15], [16]. The degraded steel regions model the effects of punching, where the degraded material proprieties depend primarily on two variables: (i) the distance from the cut edge and (ii) the magnetic field strength. There exist various approaches for modeling these effects in an FEM [17]–[20], which capture the relative decrease in permeability as a function of the distance between a considered location in the degraded region and the nearest cut-edge. Here, we are employing a relatively simple model to illustrate the UQ/SA methodology, although a more advanced model could be used if necessary. The model we utilize hinges on measurements of the steel lamination *B-H* characteristic under both degraded and non-degraded conditions, which also informs the depth of the degraded region (here, the region is 1 mm deep) [9], [16]–[18], [21], [22]. An analytic approximation that best fits the measured *B-H* data is obtained. In this study, we use a 0.3-mm thick, commercially available, non-oriented electrical steel lamination from JFE [23].

Consider the magnetization, $M(H) = B(H) - \mu_0 H$, where μ_0 is the permeability of free space. The uncertain *M-H* characteristic, in either degraded or non-degraded regions of the lamination, is modeled as a convex combination (parameterized by $\zeta_d, \zeta_{nd} \in [0, 1]$) of upper and lower bound functions,

$$M_d(H; \zeta_d) = (1 - \zeta_d) M_{d,l}(H) + \zeta_d M_{d,u}(H), \quad (1)$$

$$M_{nd}(H; \zeta_{nd}) = (1 - \zeta_{nd}) M_{nd,l}(H) + \zeta_{nd} M_{nd,u}(H), \quad (2)$$

where the subscripts *d* and *nd* stand for degraded and non-degraded, respectively, and the subscripts l and u stand for lower and upper bound, respectively. These functions are depicted in Fig. 4 for the nominal saturation value of the non-degraded upper bound function, $M^{\text{sat}}_{nd,u} = 2.12$ T.

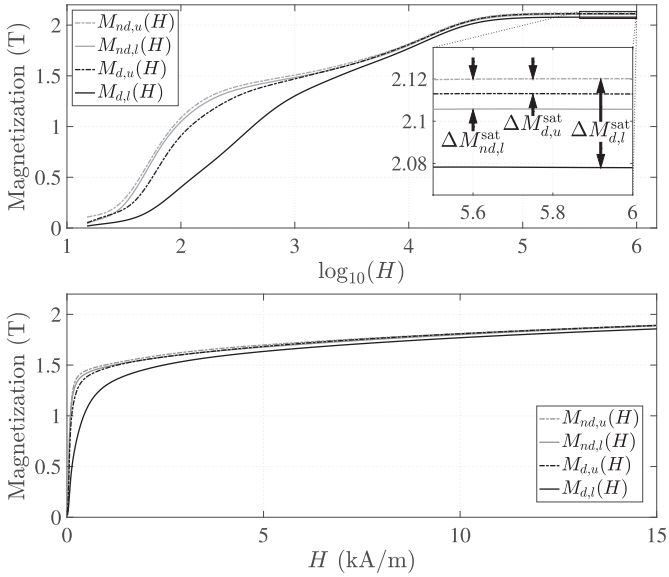


Fig. 4. Range of variation of the magnetization curves for the degraded and non-degraded regions in logarithmic and linear scales.

In order to account for the uncertainty of the saturated magnetization level (due to lack of data at sufficiently high magnetic field), we set $M^{\text{sat}}_{nd,u}$ as an uncertain independent variable. Hence, $M^{\text{sat}}_{nd,l}$, $M^{\text{sat}}_{d,u}$ and $M^{\text{sat}}_{d,l}$ are set as uncertain dependent variables such that

$$M^{\text{sat}}_{nd,l} = M^{\text{sat}}_{nd,u} - \Delta M^{\text{sat}}_{nd,l}, \quad (3)$$

$$M^{\text{sat}}_{d,u} = M^{\text{sat}}_{nd,u} - \Delta M^{\text{sat}}_{d,u}, \quad (4)$$

$$M^{\text{sat}}_{d,l} = M^{\text{sat}}_{nd,u} - \Delta M^{\text{sat}}_{d,l}, \quad (5)$$

where $\Delta M^{\text{sat}}_{nd,l} = 14.5$ mT, $\Delta M^{\text{sat}}_{d,u} = 7.2$ mT and $\Delta M^{\text{sat}}_{d,l} = 42.06$ mT are constants. These constants were selected based on the last available data points (here, at $H = 15000$ A/m). It should be noted that the extrapolation towards the saturation level is systematically achieved by the data-fitting process.

3) Parameterization of Material Loss Properties: ANSYS/Maxwell uses a time-domain approach to determine the loss of a volume of electrical steel [24]. The total core loss is separated into three types: (i) hysteresis loss, (ii) eddy current loss, and (iii) excess loss. Calculations are based on respective loss coefficients k_h, k_c, and k_e.

The variability in the material and the manufacturing process affect the loss coefficients [16], [25]. Besides, we expect the loss of the degraded steel regions to be greater than the loss of the non-degraded regions [26]. To model this effect, we define coefficient loss ratios β_h, β_c, and β_e (> 1). Consequently, for the non-degraded steel regions, we set $k_h = k_{h,nd}$, $k_c = k_{c,nd}$, and $k_e = k_{e,nd}$. On the other hand, for the degraded steel regions, we set $k_h = \beta_h k_{h,nd}$, $k_c = \beta_c k_{c,nd}$, and $k_e = \beta_e k_{e,nd}$.

4) Operating Point Parameters: The operating point of the PMSM is determined by the pair (I_s, ϕ_i), where I_s is the phase rms current, and ϕ_i is the current phase angle (assuming a

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TABLE I
INPUT PARAMETERS

Categ.	Symbol	Units	Range	Nominal ($\boldsymbol{\xi}_{d0}$)
1)	d_{or}	mm	[150.68, 150.88]	150.78
	d_{is}	mm	[151.92, 152.2]	152.06
	w_{pm}	mm	[14.447, 14.553]	14.5
	h_{pm}	mm	[6.547, 6.653]	6.6
	γ	mm	$[-0.008, 0.008]$	0
	θ_{sk}	deg	[3.2709, 4.2291]	3.75
2)	B_r	T	[1.19, 1.27]	1.23
	ζ_{nd}, ζ_d	—	[0, 1]	0.5
	$M^{\text{sat}}_{nd,u}$	T	[2.10, 2.14]	2.12
3)	$k_{h,nd}$	W/m^3	[248.68, 261.43]	255.06
	$k_{c,nd}$	W/m^3	[0.1968, 0.2068]	0.2018
	$k_{e,nd}$	W/m^3	[6.3765, 6.7035]	6.54
	$\beta_h, \beta_c, \beta_e$	—	[1, 1.5]	1.25
4)	I_s	A	[0, 424.26]	—
	ϕ_i	deg	[0, 90]	—
	$\phi_i^\star$	deg	$[-20, 110]$	—

balanced three-phase set of currents). In particular, the a-phase current is

$$i_a(\theta_r; I_s, \phi_i) = \sqrt{2} I_s \cos(\theta_r + \phi_i), \tag{6}$$

where $\theta_r = (P/2)\theta_{rm}$ is the electrical rotor angle, and P is the number of poles. To account for the skew effect, in the FEM the phase current is defined as

$$i_a^\star(\theta_r^\star; I_s, \phi_i, \theta_{sk}) = \sqrt{2} I_s \cos(\theta_r^\star + \phi_i^\star), \tag{7}$$

where the local electrical rotor angle $\theta_r^\star$ and local current phase angle $\phi_i^\star$ are defined as

$$\theta_r^\star = \theta_r + \varrho(P/2)\theta_{sk}, \quad \phi_i^\star = \phi_i - \varrho(P/2)\theta_{sk}, \tag{8}$$

with

$$\varrho = \begin{cases} 1, & \text{if counterclockwise-skewed sub-stack,} \\ -1, & \text{if clockwise-skewed sub-stack.} \end{cases} \tag{9}$$

To compute the performance of the skewed-rotor PMSM, the FEM surrogate is evaluated twice, first considering a negative skew angle ($-\theta_{sk}$, $\varrho = -1$) and then considering a positive skew angle (θ_{sk}, $\varrho = 1$), and then the outputs of each evaluation are combined accordingly.

5) Input Spaces: To summarize, let Ξ be the original input space of the skewed PMSM, and let $\boldsymbol{\xi} \in \Xi$ be an input vector (composed of 18 independent uncertain parameters)

$$\begin{aligned} \boldsymbol{\xi} = (\xi_1, \ldots, \xi_{18}) = (&d_{or}, d_{is}, w_{pm}, h_{pm}, \gamma, \\ &B_r, \zeta_{nd}, \zeta_d, M^{\text{sat}}_{nd,u}, k_{h,nd}, k_{c,nd}, k_{e,nd}, \\ &\beta_h, \beta_c, \beta_e, I_s, \phi_i, \theta_{sk}) \ \in \mathbb{R}^{18}. \end{aligned} \tag{10}$$

We also define a local input vector, $\boldsymbol{\xi}^\star \in \Xi^\star$, of the FEM as

$$\begin{aligned} \boldsymbol{\xi}^\star = (\xi_1^\star, \ldots, \xi_{17}^\star) = (&d_{or}, d_{is}, w_{pm}, h_{pm}, \gamma, \\ &B_r, \zeta_{nd}, \zeta_d, M^{\text{sat}}_{nd,u}, k_{h,nd}, k_{c,nd}, k_{e,nd}, \\ &\beta_h, \beta_c, \beta_e, I_s, \phi_i^\star) \ \in \mathbb{R}^{17}. \end{aligned} \tag{11}$$

The vectors $\boldsymbol{\xi}^\star$ and $\boldsymbol{\xi}$ are identical except for the current phase and skew angles. The parameter ranges are listed in Table I. Also, the Table shows nominal values (range mean) for the geometric and material parameters (i.e., not for the operating point). We refer to this mean value as the nominal design, and we denote it by $\boldsymbol{\xi}_{d0}$. Note that we are interested in operating points with current phase in the range $\phi_i \in [0, 90]^\circ$, considering the entire motoring operation region of the PMSM. Thus, to consider all possible scenarios of (ϕ_i, θ_{sk}) in the PMSM, the parameter $\phi_i^\star$ is allowed to vary in the slightly extended range $[-20, 110]^\circ$, based on (8).

C. FEM Output and Training Datasets

To build the surrogate model, a training dataset is defined as follows. Let $\mathcal{X} = [0, 1]^{17}$ be the 17-dimensional hypercube (i.e., the normalized input space of the FEM), and let $\mathbf{x} = (x_1, \ldots, x_{17})$ be a point in $\mathcal{X}$. The point $\mathbf{x}$ can be mapped to the FEM input space $\Xi^\star$ with

$$\xi_j^\star = \xi_{j,l}^\star + (\xi_{j,u}^\star - \xi_{j,l}^\star) x_j, \tag{12}$$

where $\xi_{j,l}^\star$ and $\xi_{j,u}^\star$ are lower and upper limits of the variability range for the j-th variable in $\Xi^\star$, respectively. To generate the training dataset, we use the Latin hypercube sampling (LHS) algorithm to draw N samples $\mathbf{x}^n$ from $\mathcal{X}$ [27]. The training input data are arranged in the matrix

$$\mathbf{X} = (\mathbf{x}^1, \ldots, \mathbf{x}^N)^\top \ \in \mathbb{R}^{N \times 17}. \tag{13}$$

The FEM calculates the electromagnetic torque waveform, the phase flux linkage waveform, and the core loss for a given input $\boldsymbol{\xi}^\star(\mathbf{x})$. The $\theta_r^\star$-dependent variables are computed at $J = 384$ different equidistant local electrical positions in the range $[0, 360)^\circ$. The periodic FEM output torque and flux linkage waveforms are

$$\boldsymbol{\tau}^\star = (\tau_1^\star, \ldots, \tau_J^\star) \in \mathbb{R}^J, \tag{14}$$

$$\boldsymbol{\psi}^\star = (\psi_1^\star, \ldots, \psi_J^\star) \in \mathbb{R}^J. \tag{15}$$

The eddy current, excess, and hysteresis losses are scalar FEM outputs, denoted by

$$p_\alpha^\star \in \mathbb{R}, \quad \alpha \in \{c, e, h\}. \tag{16}$$

The n-th sample of torque, flux linkage and loss is denoted by $\boldsymbol{\tau}^{\star n}$, $\boldsymbol{\psi}^{\star n}$, and $p_\alpha^{\star n}$, respectively.

Standardized training data are calculated for numerical stability reasons. The standardized torque waveform dataset is defined as $\{\mathbf{z}_\tau^1, \ldots, \mathbf{z}_\tau^N\}$ with

$$\mathbf{z}_\tau^n = (z_{\tau,1}^n, \ldots, z_{\tau,J}^n),\ z_{\tau,j}^n = (\tau_j^{\star n} - \mu_{\tau,j})/\sigma_{\tau,j}, \tag{17}$$

where $\mathbf{z}_\tau^n \in \mathbb{R}^J$, $\mu_{\tau,j} = \sum_{n=1}^N \tau_j^{\star n}/N$ is the mean, and $\sigma_{\tau,j}^2 = \sum_{n=1}^N (\tau_j^{\star n} - \mu_{\tau,j})^2/(N-1)$ is the variance, both over samples of torque at each local rotor position. Similarly, the standardized flux linkage waveform dataset is defined as $\{\mathbf{z}_\psi^1, \ldots, \mathbf{z}_\psi^N\}$, with

$$\mathbf{z}_\psi^n = (z_{\psi,1}^n, \ldots, z_{\psi,J}^n),\ z_{\psi,j}^n = (\psi_j^{\star n} - \mu_{\psi,j})/\sigma_{\psi,j}, \tag{18}$$

where $\mathbf{z}_\psi^n \in \mathbb{R}^J$, with mean $\mu_{\psi,j}$ and variance $\sigma_{\psi,j}^2$ over samples of flux linkage at each local rotor position. Finally, standardized eddy current, excess and hysteresis logarithm loss datasets are

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defined as

$$\mathbf{y}_\alpha = (y_\alpha^1, \ldots, y_\alpha^N)^\top,\ y_\alpha^n = (\eta_\alpha^n - \mu_\alpha)/\sigma_\alpha, \tag{19}$$

with $\alpha \in \{c, e, h\}$, $\mathbf{y}_\alpha \in \mathbb{R}^{N\times 1}$, mean μ_α, and variance σ_α^2 over samples of

$$\eta_\alpha = \log p_\alpha^\star\,. \tag{20}$$

D. Surrogate Model

Consider the general problem where we aim to map from a generic vector input $\mathbf{x}$ to the generic continuous scalar output y. Let the training dataset $\mathcal{D} \triangleq \{(\mathbf{x}^n, y^n)\}_{n=1}^N$, consisting of the training input $\mathbf{x}^n$ paired with corresponding scalar observations y^n. We wish to find a function $f(\mathbf{x})$ that predicts y for any value of $\mathbf{x}$. The computational complexity of learning this response surface in the framework of GPR is $\mathcal{O}(N^3)$ [10]. Now, consider the proposed FEM, assuming that the torque and flux linkage at each local rotor position and the core losses are independent variables. To find a scalar value function for each variable using GPR, the complexity of learning the response surface scales as $\mathcal{O}((2J+3)N^3)$. Usually, hundreds of rotor positions J are needed in order to obtain an accurate calculation of the QoIs (especially if harmonics are of interest). Therefore, building a surrogate model in this fashion can be computationally demanding.

To overcome this issue, we use PCA to reduce the dimensionality of the J-dimensional FEM outputs. PCA is a data-driven technique that has been widely used for dimensionality reduction and feature extraction [11]. Let $\mathbf{z} \in \mathbb{R}^J$ be a data point in the high-dimensional space $\mathcal{Z}$. PCA is defined as a linear projection that minimizes the distance between $\mathbf{z}$ and its approximation $\tilde{\mathbf{z}} \in \mathbb{R}^J$, given by

$$\tilde{\mathbf{z}} = \sum_{q=1}^{Q} y_q \mathbf{u}_q, \tag{21}$$

where $Q < J$ (and ideally, $Q \ll J$), and y_q is a latent variable (i.e., feature or PCA coefficient). The set of orthonormal basis vectors $\mathbf{U} = \{\mathbf{u}_1, \ldots, \mathbf{u}_Q\}$, $\mathbf{u}_q \in \mathbb{R}^J$, are the principal components (PCs), obtained from the PCA algorithm. Given $\mathbf{U}$, the linear projection of the J-dimensional point $\mathbf{z}$ onto the Q-dimensional subspace of latent variables is given by

$$y_q = \mathbf{z}^\top \mathbf{u}_q, \quad q = 1, \ldots, Q\,. \tag{22}$$

We construct a surrogate for each latent variable. To build the training dataset, we perform PCA on the standardized torque and flux linkage datasets (17) and (18) (independently of each other). Then, the training dataset observations are assembled as the vectors

$$\mathbf{y}_{\tau,q} = (y_{\tau,q}^1, \ldots, y_{\tau,q}^N)^\top \in \mathbb{R}^{N\times 1}, \quad q = 1, \ldots, Q_\tau, \tag{23}$$

$$\mathbf{y}_{\psi,q} = (y_{\psi,q}^1, \ldots, y_{\psi,q}^N)^\top \in \mathbb{R}^{N\times 1}, \quad q = 1, \ldots, Q_\psi, \tag{24}$$

containing the latent features of the torque and flux linkage, respectively. Treating the elements of (23) and (24) as mutually independent, the computational complexity of learning the response surface is reduced to $\mathcal{O}((Q_\tau + Q_\psi + 3)N^3)$. The validation error serves as the basis for determining Q_τ and Q_ψ, as described in Section II-E.

Each of the $Q_\tau + Q_\psi + 3$ surrogate functions satisfies

$$y^n = f(\mathbf{x}^n) + \varepsilon^n, \tag{25}$$

where we assume that ε^n is an additive Gaussian noise with zero mean and variance σ_y^2. The variance quantifies the discrepancy between the FEM and surrogate model outputs, but also provides numerical stability to the GPR algorithm [28]. The GP prior belief about $f(\mathbf{x})$ is described by a mean function $\mu(\mathbf{x})$ and a covariance or kernel function $k(\mathbf{x}, \mathbf{x}')$. Choosing $\mu(\mathbf{x}) = 0$, we define a joint Gaussian distribution for the vector of noisy observations $\mathbf{y} = (y^1, \ldots, y^N)^\top$, given the input dataset $\mathbf{X}$, as [10]

$$\mathbf{y}|\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{K} + \sigma_y^2 \mathbf{I}_N), \tag{26}$$

where $\mathbf{I}_N$ is the identity matrix of dimension N, and the (n, m) entry of the $N \times N$ covariance matrix $\mathbf{K}$ is given by the squared exponential kernel function,

$$\begin{aligned} K_{n,m} &= k(\mathbf{x}^n, \mathbf{x}^m; \boldsymbol{\ell}, \sigma_f) \\ &= \sigma_f^2 \exp\left\{-\frac{1}{2}(\mathbf{x}^n - \mathbf{x}^m)^\top \boldsymbol{\Lambda}^{-1}(\mathbf{x}^n - \mathbf{x}^m)\right\}, \end{aligned} \tag{27}$$

where $\boldsymbol{\Lambda} = \mathrm{diag}(\ell_1^2, \ldots, \ell_{17}^2)$. The process variance σ_f determines the average distance away from the mean function, and the characteristic length-scales $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{17})$ determine the reach of the influence of one point on its neighboring points. We use the square exponential kernel due to its smoothness feature, which yields an infinitely differentiable response surface (a property we expect to see in a PMSM model). The hyperparameters, $\boldsymbol{\nu} = (\sigma_y, \sigma_f, \boldsymbol{\ell})$, can be found by maximizing the log-marginal likelihood of $\mathbf{y}$ [10].

Then, considering a test input data point $\mathbf{x}_t$, the posterior of $f_t = f(\mathbf{x}_t)$ is, according to Bayes' rule [29],

$$f_t|\mathcal{D}, \mathbf{x}_t, \boldsymbol{\nu} \sim \mathcal{N}(\hat{\mu}(\mathbf{x}_t; \boldsymbol{\nu}), \hat{\sigma}^2(\mathbf{x}_t; \boldsymbol{\nu})), \tag{28}$$

where the predictive mean and variance are

$$\hat{\mu}(\mathbf{x}_t; \boldsymbol{\nu}) = \mathbf{k}_t^\top (\mathbf{K} + \sigma_y^2 \mathbf{I}_N)^{-1} \mathbf{y}, \tag{29}$$

$$\hat{\sigma}^2(\mathbf{x}_t; \boldsymbol{\nu}) = k(\mathbf{x}_t, \mathbf{x}_t; \boldsymbol{\ell}, \sigma_y) - \mathbf{k}_t^\top (\mathbf{K} + \sigma_y^2 \mathbf{I}_N)^{-1} \mathbf{k}_t, \tag{30}$$

respectively, with

$$\mathbf{k}_t = (k(\mathbf{x}_t, \mathbf{x}^1; \boldsymbol{\ell}, \sigma_f), \ldots, k(\mathbf{x}_t, \mathbf{x}^N; \boldsymbol{\ell}, \sigma_f))^\top\,. \tag{31}$$

Equation (29) gives the expected value of f_t, whereas the level of confidence about the prediction is described by (30). Our implementation uses the GPy framework for Python [30].

The predictive reconstructed torque and flux linkage waveforms are obtained from (17), (18), (21), and (29) as

$$\tilde{\boldsymbol{\tau}}^\star(\mathbf{x}_t) = (\tilde{\tau}_1^\star, \ldots, \tilde{\tau}_J^\star), \quad \tilde{\tau}_j^\star = \sigma_{\tau,j} \tilde{z}_{\tau,j} + \mu_{\tau,j}, \tag{32}$$

$$\tilde{\boldsymbol{\psi}}^\star(\mathbf{x}_t) = (\tilde{\psi}_1^\star, \ldots, \tilde{\psi}_J^\star), \quad \tilde{\psi}_j^\star = \sigma_{\psi,j} \tilde{z}_{\psi,j} + \mu_{\psi,j}, \tag{33}$$

where

$$\tilde{\mathbf{z}}_\tau = (\tilde{z}_{\tau,1}, \ldots, \tilde{z}_{\tau,J}) = \sum_{q=1}^{Q_\tau} \hat{\mu}_{\tau,q} \mathbf{u}_{\tau,q}, \tag{34}$$

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$$\tilde{\mathbf{z}}_\psi = (\tilde{z}_{\psi,1}, \ldots, \tilde{z}_{\psi,J}) = \sum_{q=1}^{Q_\psi} \hat{\mu}_{\psi,q} \mathbf{u}_{\psi,q} \,. \tag{35}$$

The q-th predictive latent variables related to the torque and flux linkage are denoted by $\hat{\mu}_{\tau,q}$ and $\hat{\mu}_{\psi,q}$, respectively. The predictive logarithmic core losses are obtained from (19) and (29) as

$$\tilde{\eta}_\alpha(\mathbf{x}_t) = \sigma_\alpha \hat{\mu}_\alpha + \mu_\alpha, \tag{36}$$

where $\hat{\mu}_\alpha$ is the predictive standardized logarithmic loss.

E. Validation of Surrogate Model and Computational Cost

The validation dataset is obtained by sampling N_v times the normalized FEM input space $\mathcal{X}$ using LHS, and is stored in the matrix

$$\mathbf{X}_v = (\mathbf{x}_v^1, \ldots, \mathbf{x}_v^{N_v})^\top \in \mathbb{R}^{N_v \times 17} \,. \tag{37}$$

To determine the number of PCs and to assess the accuracy of the predictive torque and flux linkage waveforms, the following validation errors are defined:

$$e_\tau = \frac{1}{N_v} \sum_{n=1}^{N_v} \frac{\|\boldsymbol{\tau}^\star(\mathbf{x}_v^n) - \tilde{\boldsymbol{\tau}}^\star(\mathbf{x}_v^n)\|}{\|\boldsymbol{\tau}^\star(\mathbf{x}_v^n)\|}, \tag{38}$$

$$e_\psi = \frac{1}{N_v} \sum_{n=1}^{N_v} \frac{\|\boldsymbol{\psi}^\star(\mathbf{x}_v^n) - \tilde{\boldsymbol{\psi}}^\star(\mathbf{x}_v^n)\|}{\|\boldsymbol{\psi}^\star(\mathbf{x}_v^n)\|}, \tag{39}$$

where $\boldsymbol{\tau}^\star(\mathbf{x}_v^n)$ and $\boldsymbol{\psi}^\star(\mathbf{x}_v^n)$ are the FEM waveform outputs at the n-th validation point, which are compared with the respective reconstructed waveform outputs $\tilde{\boldsymbol{\tau}}^\star(\mathbf{x}_v^n)$ and $\tilde{\boldsymbol{\psi}}^\star(\mathbf{x}_v^n)$ given by the surrogate, and $\|\cdot\|$ is the l^2-norm. Similarly, to assess the accuracy of the predictive logarithmic core losses we define a validation error for each type of core loss,

$$e_\alpha = \frac{1}{N_v} \sum_{n=1}^{N_v} \frac{|\eta_\alpha(\mathbf{x}_v^n) - \tilde{\eta}_\alpha(\mathbf{x}_v^n)|}{|\eta_\alpha(\mathbf{x}_v^n)|} \,. \tag{40}$$

Before performing the UQ and SA studies, we ensure that the validation errors are below a desirable threshold.

The size of the training dataset should be selected such that there is enough information to find the underlying input-output mapping over the entire input domain. However, the adequate size is not known beforehand. Thus, one sets an initial number of samples N_0, and assesses if the validation error is within the specified tolerance. If it is not the case, one repeats the training procedure by iteratively increasing the training dataset (by ΔN) until the desired accuracy is achieved.

We started with $N_0 = 500$ evaluations of the FEM, and increased the size of the training dataset by $\Delta N = 100$ in each iteration, up to 2000 FEM evaluations on the computational cluster. The size of the validation dataset was, $N_v = 200$, kept fixed through the training. We selected the first $Q_\tau = Q_\psi = 14$ PCs for the torque and flux linkage waveforms. Based on (38) and (39), the validation error was less than 0.3% at the end of the process. Note the significant dimensionality reduction, as we moved from $J = 384$ variables to $Q = 14$ features. Similarly, based on (40), we computed the validation error for the core losses, which was less than 2.1%. Fig. 5 shows the validation error as we increase the number of training samples.

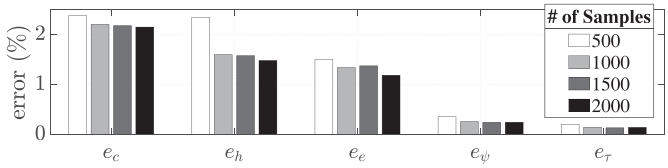


Fig. 5. Validation error of surrogate model.

TABLE II
COMPUTATIONAL COST OF TRAINING SURROGATE MODEL

# of samples	Time (s)
500	300
1000	1383
1500	3225
2000	6147

Regarding the computational cost, each FEM evaluation takes approximately 12–15 minutes, considering 384 rotor electrical positions in one electrical period. The FEM model is deployed on a computational cluster so that simulations can be parallelized. The specifications of the hardware used to run the 2200 FEM evaluations (to obtain the training and validation datasets) can be found in [31]. We used a total of 10 nodes/200 cores. With the parallelization capability of the cluster, a total computational time of approximately 48 hours was needed to carry out all the FEM evaluations. Once the training datasets are constructed, we proceed to build the $Q_\tau + Q_\psi + 3 = 31$ surrogate functions. Similarly, we used the parallelization capability of the computational cluster, with 3 nodes/60 cores. Table II summarizes the approximate total time needed to train the complete surrogate. The total time accounts for multiple restarts of the algorithm to maximize the log-marginal likelihood of the output, which is used to find the hyperparameters of the GP.

III. CASE STUDY

As a case study, we select 4 QoIs, namely, (i) the average torque $\tau_{\text{avg}}(\boldsymbol{\xi})$; (ii) the sixth torque harmonic magnitude $\tau_{\text{6th}}(\boldsymbol{\xi})$, which affects the PMSM noise and vibration; (iii) the peak line-to-line voltage $V_{pk}(\boldsymbol{\xi}, \omega_r) = \max(|v_a(\theta_r; \boldsymbol{\xi}, \omega_r) - v_b(\theta_r; \boldsymbol{\xi}, \omega_r)|)$, which determines the dc-link voltage; and (iv) the total core loss $p_t(\boldsymbol{\xi}, \omega_r)$. The QoIs are computed by post-processing the surrogate outputs based on Fourier series of the skewed PMSM electromagnetic torque, $\tau(\theta_r; \boldsymbol{\xi})$, and a-phase flux linkage, $\psi_a(\theta_r; \boldsymbol{\xi})$. The a-phase voltage is [32]

$$v_a(\theta_r; \boldsymbol{\xi}, \omega_r) = R_s i_a(\theta_r; I_s, \phi_i) + \omega_r \left[L_e \frac{d i_a(\theta_r; I_s, \phi_i)}{d\theta_r} + \frac{d\psi_a(\theta_r; \boldsymbol{\xi})}{d\theta_r} \right], \tag{41}$$

where $\omega_r = d\theta_r/dt$ is the electrical angular velocity, $R_s = 12.16\ \text{m}\Omega$ is the phase resistance, and $L_e = 10\ \mu\text{H}$ is the end-turn leakage inductance. The FEM and, hence, the surrogate model

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yield the core loss for a fixed angular electrical speed $\omega_r^\star$. We evaluate the total core loss at arbitrary ω_r with

$$p_t(\boldsymbol{\xi}, \omega_r) = \lambda^2 p_c(\boldsymbol{\xi}, \omega_r^\star) + \lambda^{1.5} p_e(\boldsymbol{\xi}, \omega_r^\star) + \lambda p_h(\boldsymbol{\xi}, \omega_r^\star), \quad (42)$$

where $p_c(\boldsymbol{\xi}, \omega_r^\star)$, $p_e(\boldsymbol{\xi}, \omega_r^\star)$, and $p_h(\boldsymbol{\xi}, \omega_r^\star)$ are the eddy current, excess and hysteresis losses of the skewed PMSM at $\omega_r^\star$, respectively, and $\lambda = \omega_r / \omega_r^\star$.

Although we trained the surrogates assuming input parameters are independent, in this case study we introduce a simple correlation between the degradation of the magnetic properties and the loss coefficients within degraded regions [16], [25]. Therefore, we assume $\beta_c = \beta_e = \beta_h = \beta_\alpha(\zeta_d)$, with

$$\beta_\alpha(\zeta_d) = \beta_{\alpha,l} + (\beta_{\alpha,u} - \beta_{\alpha,l})(1 - \zeta_d), \quad (43)$$

so that maximum degradation ($\zeta_d = 0$) coincides with the maximum loss ratio.

Our case study considers a notional PMSM drive operating under a maximum torque per watt (MTPW) control strategy. The drive is programmed based on the nominal motor ($\boldsymbol{\xi}_{d0}$). An optimization problem is thus posed to determine the current command (I_s', ϕ_i') that will generate a desired average torque τ_{avg}' for a given rotor speed ω_r',

$$\begin{aligned} \min_{I_s, \phi_i} \quad & 3R_s I_s^2 + p_t(I_s, \phi_i, \boldsymbol{\xi}_{d0}, \omega_r') \\ \text{subject to} \quad & \tau_{\text{avg}}(I_s, \phi_i, \boldsymbol{\xi}_{d0}) \geq \tau_{\text{avg}}' \\ & V_{pk}(I_s, \phi_i, \boldsymbol{\xi}_{d0}, \omega_r') \leq V_{\text{dc}} \\ & I_s \leq I_{\max}, \end{aligned} \quad (44)$$

where $V_{\text{dc}} = 420$ V is the dc-link voltage of the inverter driving the PMSM, and $I_{\max} = 353.5$ A is the maximum stator current rms value. We solve this problem at discrete points of τ_{avg}' in the range [25, 230] Nm, and for $\omega_r = 1633.6$ rad/s, which is the rated speed of the nominal PMSM. Note that $\tau_{\text{avg}}(I_s, \phi_i, \boldsymbol{\xi}_{d0})$, $V_{pk}(I_s, \phi_i, \boldsymbol{\xi}_{d0}, \omega_r')$, and $p_t(I_s, \phi_i, \boldsymbol{\xi}_{d0}, \omega_r')$ are computed based on (32)–(36) (accounting for all sub-stacks, accordingly). Fig. 6 depicts the MTPW current commands (I_s', ϕ_i') as a function of torque command τ_{avg}'. Also, it depicts the nominal values of the remaining three QoIs, $\tau_{\text{6th}}' = \tau_{\text{6th}}(I_s', \phi_i', \boldsymbol{\xi}_{d0})$, $V_{pk}' = V_{pk}(I_s', \phi_i', \boldsymbol{\xi}_{d0}, \omega_r')$, and $p_t' = p_t(I_s', \phi_i', \boldsymbol{\xi}_{d0}, \omega_r')$.

We perform forward UQ and SA studies to quantify the effects and relevance of the uncertain parameters on the QoIs. For this particular PMSM, we analyze the deviation of the actual from the expected operation due to the manufacturing variability. The geometric and material parameters of an off-nominal machine are denoted as $\boldsymbol{\xi}_d$. We define deviations from nominal operation as

$$\Delta\tau_{\text{avg}} = \tau_{\text{avg}}' - \tau_{\text{avg}}(I_s', \phi_i', \boldsymbol{\xi}_d), \quad (45)$$

$$\Delta\tau_{\text{6th}} = \tau_{\text{6th}}' - \tau_{\text{6th}}(I_s', \phi_i', \boldsymbol{\xi}_d), \quad (46)$$

$$\Delta V_{pk} = V_{pk}' - V_{pk}(I_s', \phi_i', \boldsymbol{\xi}_d, \omega_r'), \quad (47)$$

$$\Delta p_t = p_t' - p_t(I_s', \phi_i', \boldsymbol{\xi}_d, \omega_r'). \quad (48)$$

We assume that $\boldsymbol{\xi}_d$ is uniformly distributed within the ranges of Table I. To conduct a forward UQ study, a Monte Carlo approach is followed, where the uncertainty in $\boldsymbol{\xi}_d$ is propagated through the surrogate model, intending to estimate the induced statistics of the QoIs. On the other hand, to conduct an SA study, a variance-based approach is followed, where we use Monte Carlo-based estimators for the computation of first-order sensitivity indices, also known as first-order Sobol indices [33]. This measure of sensitivity determines the relative relevance of the parameters to the model response, where values closer to one indicate higher relevance. We use the Sensitivity Analysis Library (SALib) in Python to estimate the Sobol indices [34]. Both UQ and SA are conducted based on 28000 random samples $\boldsymbol{\xi}_d$ at each MTPW operating point.

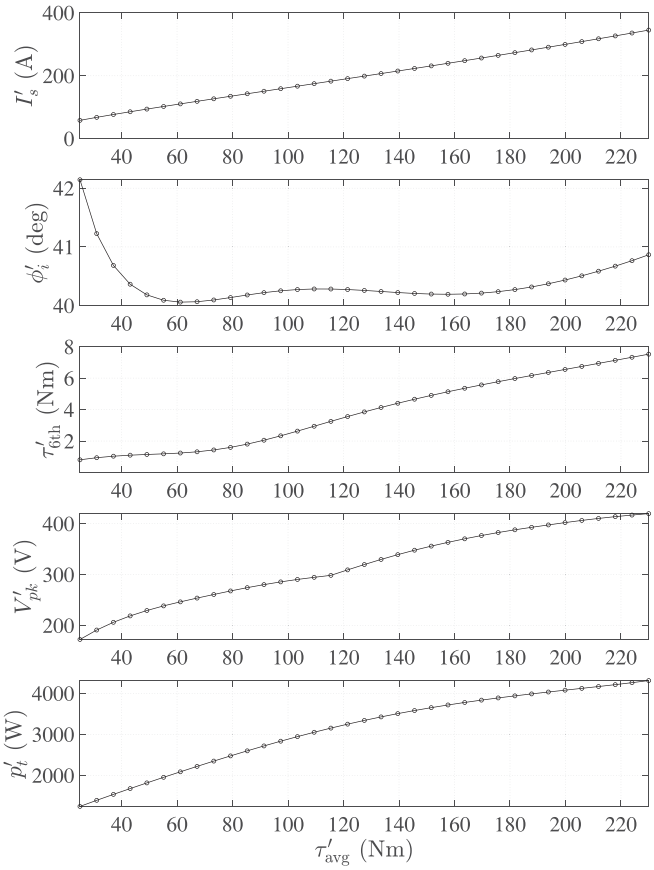


Fig. 6. Predictive PMSM nominal operation.

Under the assumption that the input is uniformly distributed, for the PMSM under study, Fig. 7 shows the deviation of actual from expected operation due to the manufacturing variability as a function of the torque command. The respective 99% central credible intervals are plotted as grey zones. For example, at rated speed with $\tau_{\text{avg}}' = 230$ Nm, the results suggest that the expected QoI values are (with 99% confidence): $\tau_{\text{avg}} \in [219.12,\ 240.29]$ Nm, $\tau_{\text{6th}} \in [6.36,\ 8.8]$ Nm, $V_{pk} \in [405.19,\ 433.46]$ V, and $p_t = [3.13,\ 5.57]$ kW. In addition, Fig. 7 also shows the Sobol indices of the most relevant parameters affecting each QoI as a function of the torque command.

Regarding the average torque, the results suggest that the uncertainty about its value increases as the torque command increases. Based on the Sobol indices, the stator inner diameter

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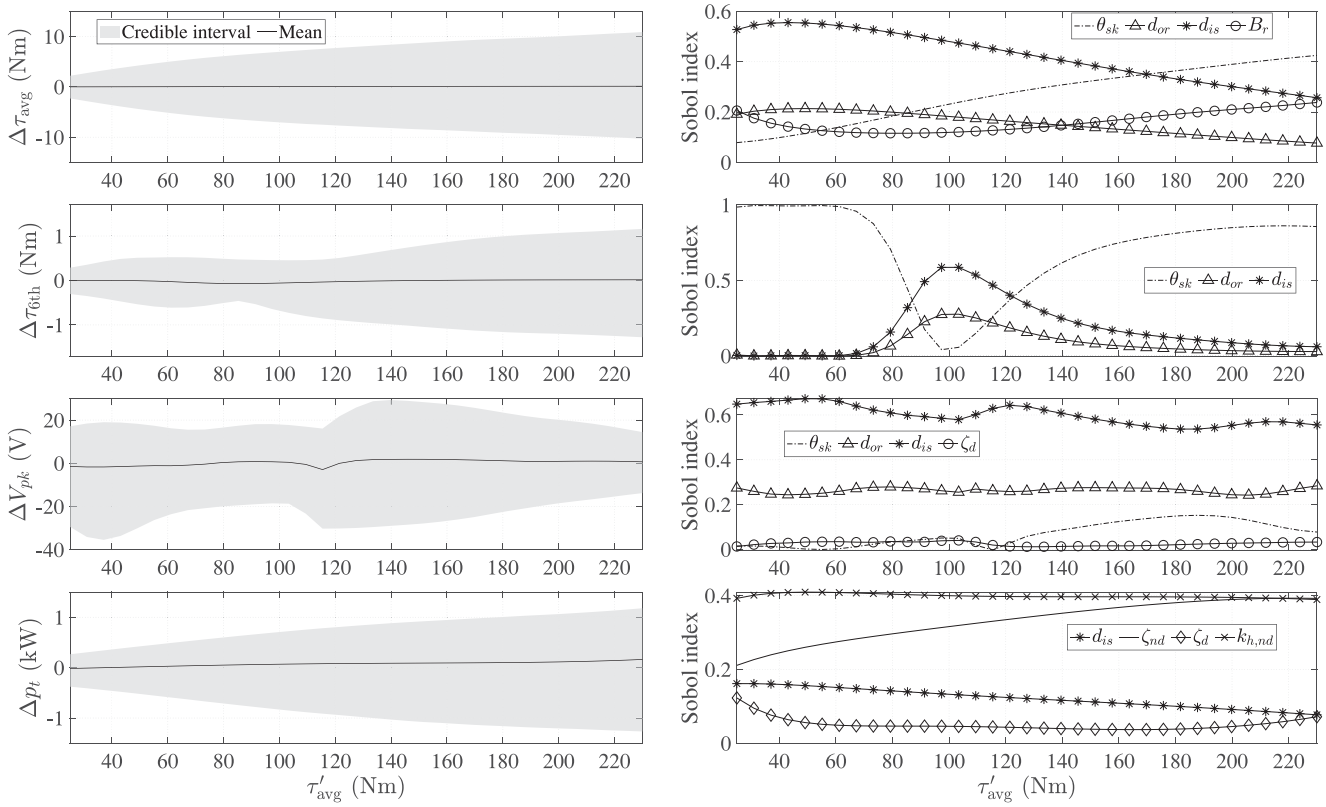


Fig. 7. UQ and SA results.

d_{is} explains the major variability in the average torque up to $\tau'_{\text{avg}} = 165$ Nm (roughly 38–58% of the variability), whereas for $\tau'_{\text{avg}} > 165$ Nm, the skew angle becomes the most relevant parameter (explaining at least 38% of the variability). Note that the relevance of the skew angle increases as the torque command increases, while the relevance of the stator inner diameter decreases. It is evident that the most relevant parameter does not only depend on manufacturing imperfections, but it is also operating point-dependent. This could be used as an indicator of which part of the manufacturing process should be targeted to reduce the uncertainty in the average torque, taking into account the operating conditions of the PMSM. The magnet remanence and the outer rotor diameter are the subsequent most relevant parameters, explaining in conjunction approximately 30% of the variability in the average torque over the entire range of τ'_{avg}. Likewise, we can conclude that the pocket and magnet dimensions do not affect the average torque for the considered ranges of variability in this PMSM. Lastly, the results suggest that the variability of the magnetic properties of steel does not significantly affect the deviation of actual from expected average torque.

Concerning the sixth torque harmonic magnitude, we can see that the uncertainty about τ_{6th} mostly increases as the torque command increases (except between $\tau'_{\text{avg}} = 60$ Nm and $\tau'_{\text{avg}} = 90$ Nm, where a small decrease is depicted). However, the largest deviation of the actual from the expected value is less than 1.3 Nm. Here, we can observe that the skew angle explains 100% of the uncertainty up to $\tau'_{\text{avg}} = 60$ Nm. Then, an abrupt change in the Sobol indices indicates that around 100 Nm the most relevant parameters are the stator inner diameter and the rotor outer diameter, that in conjunction explain about 80% of the uncertainty in τ_{6th}. For $\tau'_{\text{avg}} > 120$ Nm, the variability in the skew angle becomes again the most significant cause of uncertainty. Similar to τ_{avg}, the variability of material properties of steel and magnets, and the geometric features of the magnets and pockets do not impact τ_{6th}.

Regarding the peak line-to-line voltage, the results suggest that the uncertainty about the deviation of actual from expected varies nonlinearly with the torque command. For rated speed, with $\tau'_{\text{avg}} = 37$ Nm, the maximum deviation is 35.5 V, while with $\tau'_{\text{avg}} = 230$ Nm, the maximum deviation is 14.5 V. Similarly, d_{is} is the most significant factor for the variability of V_{pk}. The stator inner diameter explains, on average, about 60% of the uncertainty throughout the entire range of τ'_{avg}. Most of the other 40% is explained in conjunction by θ_{sk}, d_{or} and ζ_d. Again, the variability of material properties and the geometric features of the magnets and pockets do not significantly impact the peak line-to-line voltage. However, there is a small influence ($<5\%$) of ζ_d over the uncertainty of V_{pk}.

Finally, the non-degraded hysteresis loss coefficient explains 40% of the variability in the PMSM core loss. The subsequent most relevant parameter is ζ_{nd}. Its relevance increases as the torque command increases (changing from 20% to 40%). The rest of the uncertainty is caused by d_{is} and ζ_d. Besides, the results suggest that the uncertainty about p_t increases as the torque command increases.

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This analysis suggests that reducing the uncertainty about the inner stator diameter, the skew angle, and the non-degraded hysteresis loss coefficient could substantially reduce the uncertainty in the QoIs.

IV. Conclusion

We have set forth a computationally efficient methodology to quantify the effects and relevance of mass-production variability on a PMSM. The proposed method can simultaneously account for tens of uncertain variables related to geometric and material uncertainty. The fundamental physical quantities of interest, namely, the torque and flux linkage waveforms and the core loss, are extracted from a nonlinear finite element model. Then, an FEM surrogate is constructed using Gaussian process regression. A novel method that hinges on principal component analysis for the waveforms is proposed to reduce the problem dimensionality.

Subsequently, we demonstrate how the surrogate models can be used to conduct uncertainty quantification and sensitivity analysis studies to determine the level of confidence about the PMSM performance, and the relative importance of the parameters. The results provide a practical indication about which stage of the mass manufacturing should be targeted to increase robustness.

Our methodology is necessarily restricted to a few tens of uncertain variables as the number of simulations required to build an accurate surrogate scales exponentially with increasing number of inputs (curse of dimensionality). One can overcome this issue only if the response exhibits some special structure that can be identified and exploited to reduce the dimensionality of the input. See [35] for an example of linear input dimensionality reduction in the context of Gaussian process regression and [35], [36] for an example of nonlinear input dimensionality reduction using deep neural networks. In our future work, we are considering expanding this methodology to scenarios of interest including operational uncertainty and thermal effects, as well as for the robust design of electric motors.

References

[1] M. S. Islam, S. Mir, and T. Sebastian, "Issues in reducing the cogging torque of mass-produced permanent-magnet brushless DC motor," *IEEE Trans. Ind. Appl.*, vol. 40, no. 3, pp. 813–820, May 2004.

[2] L. Gašparin, A. Černigoj, S. Markič, and R. Fišer, "Additional cogging torque components in permanent-magnet motors due to manufacturing imperfections," *IEEE Trans. Magn.*, vol. 45, no. 3, pp. 1210–1213, Mar. 2009.

[3] K. Bourchas *et al.*, "Quantifying effects of cutting and welding on magnetic properties of electrical steels," *IEEE Trans. Ind. Appl.*, vol. 53, no. 5, pp. 4269–4278, Sep. 2017.

[4] A. Al-Timimy, G. Vakil, M. Degano, P. Giangrande, C. Gerada, and M. Galea, "Considerations on the effects that core material machining has on an electrical machine's performance," *IEEE Trans. Energy Convers.*, vol. 33, no. 3, pp. 1154–1163, Sep. 2018.

[5] R. C. Smith, *Uncertainty Quantification: Theory, Implementation, and Applications*. Philadelphia, PA, USA: SIAM, 2013.

[6] A. Saltelli *et al.*, *Global Sensitivity Analysis. The Primer*. Piscataway, NJ, USA: Wiley, 2008.

[7] C. P. Robert and G. Casella, *Monte Carlo Statistical Methods*. New York, NY, USA: Springer, 2004.

[8] H. M. S. Harstick, M. Ritter, and W. Riehemann, "Influence of punching and tool wear on the magnetic properties of non-oriented electrical steel," *IEEE Trans. Magn.*, vol. 50, no. 4, pp. 1–4, Apr. 2014.

[9] M. Bali and A. Muetze, "The degradation depth of non-grain oriented electrical steel sheets of electric machines due to mechanical and laser cutting: A state-of-the-art review," *IEEE Trans. Ind. Appl.*, vol. 55, no. 1, pp. 366–375, Jan. 2019.

[10] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT Press, 2005.

[11] C. M. Bishop, *Pattern Recognition and Machine Learning*, 1st ed. New York, NY, USA: Springer, 2006.

[12] ANSYS Inc., "ANSYS Maxwell," 2020. [Online]. Available: https://www.ansys.com/products/electronics/ansys-maxwell, Accessed: Dec. 2019.

[13] I. Coenen, M. van der Giet, and K. Hameyer, "Manufacturing tolerances: Estimation and prediction of cogging torque influenced by magnetization faults," *IEEE Trans. Magn.*, vol. 48, no. 5, pp. 1932–1936, May 2012.

[14] A. Kallaste, A. Belahcen, and T. Vaimann, "Effect of PM parameters variability on the operation quantities of a wind generator," in *Proc. IEEE Workshop Elect. Mach. Des., Control Diagnosis*, Mar. 2015, pp. 242–247.

[15] O. Fischer and J. Schneider, "Influence of deformation process on the improvement of non-oriented electrical steel," *J. Magn. Magn. Mater.*, vol. 254-255, pp. 302–306, Jan. 2003.

[16] A. Krings, "Iron losses in electrical machines: Influence of material properties, manufacturing processes and inverter operation," Ph.D. dissertation, KTH Roy. Institute Technol., Stockholm, Sweden, May 2014.

[17] M. Bali, H. De Gersem, and A. Muetze, "Finite-element modeling of magnetic material degradation due to punching," *IEEE Trans. Magn.*, vol. 50, no. 2, pp. 745–748, Feb. 2014.

[18] H. Sano, K. Narita, E. Zeze, T. Yamada, U. Kazuki, and K. Akatsu, "A practical approach for electromagnetic analysis with the effect of the residual strain due to manufacturing processes," in *Proc. IEEE Energy Convers. Cong. Expo.*, Sep. 2016, pp. 1–7.

[19] R. Sundaria, A. Lehikoinen, A. Hannukainen, A. Arkkio, and A. Belahcen, "Mixed-order finite-element modeling of magnetic material degradation due to cutting," *IEEE Trans. Magn.*, vol. 54, no. 6, pp. 1–8, Jun. 2018.

[20] N. M'zali, F. Martin, R. Sundaria, T. Henneron, A. Benabou, and A. Belahcen, "Finite-Element modeling of magnetic properties degradation due to plastic deformation," *IEEE Trans. Magn.*, vol. 56, no. 2, pp. 1–4, Feb. 2020.

[21] A. J. Moses, N. Derebasi, G. Loisos, and A. Schoppa, "Aspects of the cut-edge effect stress on the power loss and flux density distribution in electrical steel sheets," *J. Magn. Magn. Mater.*, vol. 215, pp. 690–692, Jun. 2000.

[22] F. Ossart, E. Hug, O. Hubert, C. Buvat, and R. Billardon, "Effect of punching on electrical steels: Experimental and numerical coupled analysis," *IEEE Trans. Magn.*, vol. 36, no. 5, pp. 3137–3140, Sep. 2000.

[23] JFE Steel Corporation, "JFE," 2020. [Online]. Available: https://www.jfe-steel.co.jp, Accessed: Apr. 2020.

[24] D. Lin, P. Zhou, W. N. Fu, Z. Badics, and Z. J. Cendes, "A dynamic core loss model for soft ferromagnetic and power ferrite materials in transient finite element analysis," *IEEE Trans. Magn.*, vol. 40, no. 2, pp. 1318–1321, Mar. 2004.

[25] R. Ramarotafika, A. Benabou, S. Clenet, and J. C. Mipo, "Experimental characterization of the iron losses variability in stators of electrical machines," *IEEE Trans. Magn.*, vol. 48, no. 4, pp. 1629–1632, Apr. 2012.

[26] K. Bourchas *et al.*, "Quantifying effects of cutting and welding on magnetic properties of electrical steels," *IEEE Trans. Ind. Appl.*, vol. 53, no. 5, pp. 4269–4278, Sep. 2017.

[27] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, May 1979.

[28] R. Ababou, A. C. Bagtzoglou, and E. F. Wood, "On the condition number of covariance matrices in kriging, estimation, and simulation of random fields," *Math. Geol.*, vol. 26, no. 1, pp. 99–133, Jan. 1994.

[29] I. Bilionis, N. Zabaras, B. A. Konomi, and G. Lin, "Multi-output separable Gaussian process: Towards an efficient, fully Bayesian paradigm for uncertainty quantification," *J. Comput. Phys.*, vol. 241, pp. 212–239, May 2013.

[30] GPy, "GPy: A Gaussian processes framework in Python," 2012. [Online]. Available: http://github.com/SheffieldML/GPy

[31] Information Technology at Purdue, "Rice community cluster," 2020. [Online]. Available: https://www.rcac.purdue.edu/compute/rice/, Accessed: May, 2020.

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[32] P. C. Krause, O. Wasynczuk, S. Sudhoff, and S. Pekarek, *Analysis of Electric Machinery and Drive Systems*, 3rd ed. Piscataway, NJ, USA: Wiley, 2013.
[33] I. M. Sobol, "Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates," *Math. Comput. Simul.*, vol. 55, no. 1–3, pp. 271–280, Feb. 2001.
[34] J. Herman and W. Usher, "SALib: An open-source Python library for sensitivity analysis," *J. Open Source Softw.*, vol. 2, no. 9, Jan. 2017.
[35] R. Tripathy, I. Bilionis, and M. Gonzalez, "Gaussian processes with built-in dimensionality reduction: Applications to high-dimensional uncertainty propagation," *J. Comput. Phys.*, vol. 321, pp. 191–223, Sep. 2016.
[36] R. Tripathy and I. Bilionis, "Deep UQ: Learning deep neural network surrogate models for high dimensional uncertainty quantification," *J. Comput. Phys.*, vol. 375, pp. 565–588, Dec. 2018.

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